

Name:

INTRODUCTION TO CRYPTOGRAPHY & SECURITY

Student ID:

Exam duration: 60 minutes.

Multiple-Choice Questions

1. What is the most appropriate cryptographic primitive to use if a company wants to distribute authenticated software updates to its customers?
 - A. Hash function.
 - B. Digital signature scheme.
 - C. Message authentication code.
 - D. Block cipher
 - E. Public-key encryption scheme.
2. What is the most appropriate cryptographic primitive to use if an individual wants to ensure confidentiality of the files stored on her hard drive?
 - A. Message authentication code. for integrity not for confidentiality
 - B. Block cipher.
 - C. Public-key encryption.
 - D. Hash function. for integrity not for confidentiality
 - E. Private-key encryption.
3. Which of the following best characterizes Discretionary Access Control (DAC)?
 - A. Access decisions based on security labels.
 - B. An entity can grant access to others.
 - C. Centralized control by a security administrator.
 - D. Access based on roles within an organization.
4. Consider the following authentication protocol between Alice and Bob:
 1. Bob $\rightarrow$ Alice: n_B
 2. Alice $\rightarrow$ Bob: $n_A \parallel \text{sig}_A(n_A \parallel n_B \parallel ID_B)$
 3. Bob $\rightarrow$ Alice: $n'_B \parallel \text{sig}_B(n'_B \parallel n_A \parallel ID_A)$

What is the primary objective of Alice's message in step 2?

- A. To send her nonce n_A to Bob.
- B. To prove her identity to Bob and acknowledge Bob's nonce n_B .
- C. To establish a shared secret key with Bob.
- D. To request Bob's public key.

5. Let X be a uniform random variable over the set $\{0, 1\}^n$. Let Y be an arbitrary random variable over the set $\{0, 1\}^n$ (not necessarily uniform) that is independent of X .

Define the random variable $Z = X \oplus Y$. What is the probability that Z equals 0^n ?

- A. 0.5
 - B. $1/2^n$
 - C. $2/2^n$
 - D. 0
6. Which of the following numbers is a generator of $\mathbb{Z}_{17}^*$?
- A. 2, $\langle 2 \rangle = \{1, 2, 4, 8, 16, 15, 13, 9, 1, 2, 4, 8, 16, 15, 13, 9\}$
 - B. 3, $\langle 3 \rangle = \{1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6\}$
 - C. 5, $\langle 5 \rangle = \{1, 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7\}$
 - D. 7, $\langle 7 \rangle = \{1, 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5\}$
 - E. 6, $\langle 6 \rangle = \{1, 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3\}$
 - F. 8, $\langle 8 \rangle = \{1, 8, 13, 2, 16, 9, 4, 15, 1, 8, 13, 2, 16, 9, 4, 15\}$
 - G. 9, $\langle 9 \rangle = \{1, 9, 13, 15, 16, 8, 4, 2, 1, 9, 13, 15, 16, 8, 4, 2\}$
 - H. 10, $\langle 10 \rangle = \{1, 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12\}$
7. Let G be a finite cyclic group of order q and g is a generator of G . Suppose that the discrete log problem is hard in G . Which of the following problems is also hard:
- A. Randomly pick $y \in G$, find x such that $g^x = y$.
 - B. Randomly pick two values $x \in \mathbb{Z}_q$ and $y \in G$, compute $y^x \cdot g$.
 - C. Randomly pick $x \in \mathbb{Z}_q$, find y such that $g^x = y$
 - D. Find x and y such that $g^x = y$.
 - E. Randomly pick $y \in G$, find x such that $g^{x+1} = y$.
8. In the following problems, N is the product of two large integers p and q ; and $e < N$ such that $\gcd(e, \varphi(N)) = 1$.
- If RSA is hard, which of the following problems is hard?
- A. Given N , e , and randomly pick $y \in \mathbb{Z}_N^*$, find x such that $x^e = y \bmod N$.
 - B. Given N and e , find x, y such that $x^e = y \bmod N$.
 - C. Given N and e , find x such that $x^e = 8 \bmod N$.
 - D. Given N, e , and randomly pick $x \in \mathbb{Z}_N^*$, find y such that $x^e = y \bmod N$
9. Which of the following statements is true about the basic Diffie-Hellman key-exchange protocol.

- A. The basic protocol provides key exchange secure against active adversaries that can inject and modify messages.
 - B. As with RSA, the protocol only provides eavesdropping security in the group $\mathbb{Z}_N^*$ where N is an RSA modulus.
 - C. The protocol can be converted to a public-key encryption system called the ElGamal public-key system.
10. Let $H : M \rightarrow T$ be a collision resistant hash function where $|T|$ is smaller than $|M|$. Which of the following properties is implied by collision resistance?
- A. Given $H(m)$ the attacker learns nothing about m .
 - B. It is difficult to find m_0 and m_1 such that $H(m_0) = H(m_1) + 1$. (here we treat the outputs of H as integers)
 - C. For all m in M , $H(m)$ must be shorter than m .
 - D. It is difficult to construct two distinct messages m_0, m_1 such that $H(m_0) = H(m_1)$.
11. Let (S, V) be a secure MAC defined over (K, M, T) where $M = \{0, 1\}^n$ and $T = \{0, 1\}^{128}$ (i.e. the key space is K , message space is $\{0, 1\}^n$, and tag space is $\{0, 1\}^{128}$). Which of the following is a secure MAC: (as usual, we use $\parallel$ to denote string concatenation).

A.

$$S'(k, m) = S(k, m[0, \dots, n-2] \parallel 0)$$

$$V'(k, m, t) = V(k, m[0, \dots, n-2] \parallel 0, t)$$

B.

$$S'(k, m) = S(k, m)[0, \dots, 126]$$

$$V'(k, m, t) = [V(k, m, t \parallel 0) \text{ or } V(k, m, t \parallel 1)]$$

(i.e., $V'(k, m, t)$ outputs “1” if either $t \parallel 0$ or $t \parallel 1$ is a valid tag for m).

C.

$$S'(k, m) = S(k, m \parallel m)$$

$$V'(k, m, t) = V(k, m \parallel m, t).$$

D.

$$S'(k, m) = S(k, m)$$

$$V'(k, m, t) = \begin{cases} V(k, m, t) & \text{if } m \neq 0^n \\ \text{“1”} & \text{otherwise} \end{cases}$$

E.

$$S'(k, m) = S(k, m) \quad \text{and} \\ V'(k, m, t) = [V(k, m, t) \text{ or } V(k, m \oplus 1^n, t)]$$

(i.e., $V'(k, m, t)$ outputs “1” if t is a valid tag for either m or $m \oplus 1^n$).

F.

$$S'((k_1, k_2), m) = (S(k_1, m), S(k_2, m)) \quad \text{and} \\ V'((k_1, k_2), m, (t_1, t_2)) = [V(k_1, m, t_1) \text{ and } V(k_2, m, t_2)]$$

(i.e., $V'((k_1, k_2), m, (t_1, t_2))$ outputs “1” if both t_1 and t_2 are valid tags).

12. Suppose a MAC system (S, V) is used to protect files in a file system by appending a MAC tag to each file. The MAC signing algorithm S is applied to the file contents and nothing else. What tampering attacks are not prevented by this system?
- A. Swapping two files in the file system.
 - B. Replacing the tag and contents of one file with the tag and contents of a file from another computer protected by the same MAC system, but a different key.
 - C. Replacing the contents of a file with the concatenation of two files on the file system.
 - D. Erasing the last byte of the file contents.
13. Given an RSA signature scheme with the public key $(n = 9797, e = 7)$, which of the following signatures are valid?
- A. $(m = 2601, \sigma = 7496)$
 - B. $(m = 1894, \sigma = 8667)$
 - C. $(m = 4707, \sigma = 4976)$
 - D. $(m = 2016, \sigma = 8159)$

Calculation Questions

14. Suppose you are told that the one time pad encryption of the message “gui duc 100d” is

5c73d5340a948c86981bc294

(the plaintext letters are encoded as 8-bit ASCII and the given ciphertext is written in hex). The one time pad encryption of the message “gui duc 211d” under the same OTP key is _____.

Note: you just write the 8-bit ASCII code of the ciphertext in hexadecimal.

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15. Using Table 1, the inverse of the bytes 29 is _____.

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X	Y															
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	01	8D	F6	CB	52	7B	D1	E8	4F	29	C0	B0	E1	E5	C7
1	74	B4	AA	4B	99	2B	60	5F	58	3F	FD	CC	FF	40	EE	B2
2	3A	6E	5A	F1	55	4D	A8	C9	C1	0A	98	15	30	44	A2	C2
3	2C	45	92	6C	F3	39	66	42	F2	35	20	6F	77	BB	59	19
4	1D	FE	37	67	2D	31	F5	69	A7	64	AB	13	54	25	E9	09
5	ED	5C	05	CA	4C	24	87	BF	18	3E	22	F0	51	EC	61	17
6	16	5E	AF	D3	49	A6	36	43	F4	47	91	DF	33	93	21	3B
7	79	B7	97	85	10	B5	BA	3C	B6	70	D0	06	A1	FA	81	82
8	83	7E	7F	80	96	73	BE	56	9B	9E	95	D9	F7	02	B9	A4
9	DE	6A	32	6D	D8	8A	84	72	2A	14	9F	88	F9	DC	89	9A
A	FB	7C	2E	C3	8F	B8	65	48	26	C8	12	4A	CE	E7	D2	62
B	0C	E0	1F	EF	11	75	78	71	A5	8E	76	3D	BD	BC	86	57
C	0B	28	2F	A3	DA	D4	E4	0F	A9	27	53	04	1B	FC	AC	E6
D	7A	07	AE	63	C5	DB	E2	EA	94	8B	C4	D5	9D	F8	90	6B
E	B1	0D	D6	EB	C6	0E	CF	AD	08	4E	D7	E3	5D	50	1E	B3
F	5B	23	38	34	68	46	03	8C	DD	9C	7D	A0	CD	1A	41	1C

Table 1: Multiplicative Inverses in $\text{GF}(2^8)$ for AES S-box

16. The BkExam is a block cipher optimized for use on final exam. It operates on the 26 letters of the English alphabet:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
m	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$E_K(m)$	P	K	X	C	Y	W	R	S	E	J	U	D	G	O	Z	A	T	N	M	V	F	H	L	I	B	Q

As the $\oplus$ operation is not defined on the set $\{A, \dots, Z\}$, we replace it here during encryption with modulo-26 addition (e.g., $C \oplus D = F$ and $Y \oplus C = A$).

The ciphertext of the message: “DHBK” which were encrypted using **CBC mode** with $IV=H$ is _____.

Note: You should add IV to the ciphertext.

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17. Let m be a message consisting of ℓ AES blocks (say $\ell = 100$). Alice encrypts m using randomized counter mode and transmits the resulting ciphertext to Bob. Due to a network error, ciphertext block number $\ell/2$ is corrupted during transmission. All other ciphertext blocks are transmitted and received correctly. Once Bob decrypts the received ciphertext, how many plaintext blocks will be corrupted? _____

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18. Compute $30^{-1} \bmod 113$.

The number $30^{-1} \bmod 113$ is _____.

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19. Compute $779^{168002} \bmod 11413$ knowing that $11413 = (101 \times 113)$.

The number $779^{168002} \bmod 11413$ is _____.

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20. What is the discrete log of 4 base 3 in $\mathbb{Z}_{17}^*$.

The number $\text{Dlog}_3(4)$ is _____.

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21. An RSA encryption scheme has the set-up parameters $p = 13$ and $q = 19$. The public key is $e = 17$. You are asked to decrypt the ciphertext $y = 16$.

The plaintext x is _____.

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22. Let $\mathbb{Z}_{19}^*$ be a cyclic group and 3 is a generator. Compute $\text{DH}_3(12, 15)$ in this group.

The value $\text{DH}_3(12, 15)$ is _____.

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23. Given an elliptic curve

$$E : \quad y^2 = x^3 + 2x + 2 \pmod{17}$$

The base point $P = (13, 10)$. Compute the point $Q = 78 \cdot P$.

The point Q is _____.

Note: Write the result as (p,q) with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

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24. Given an elliptic curve

$$E : \quad y^2 = x^3 + 2x + 2 \pmod{17}$$

Consider the point $P = (13, 10)$. Alice and Bob are to compute a shared key in a Diffie-Hellman protocol based on elliptic curve E . That is,

- Alice randomly pick a number a and compute the point $aP = (16, 13)$, and send to Bob;
- Alice receive the point $bP = (16, 13)$ from Bob.

The shared key abP between Alice and Bob is _____.

Note: Write the result as **(p,q)** with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

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25. We consider the ElGamal signature scheme. You are given Bob's private key $sk = d = (12)$ and the corresponding public key $pk = (p, g, g^d) = (29, 2, 7)$. Compute the Elgamal signature (r, s) for a message from Bob to Alice with message $m = 20$ and ephemeral key $k_E = 5$.

The signature (r, s) is _____.

Note: Write the signature as **(r,s)** with no spaces in between.

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